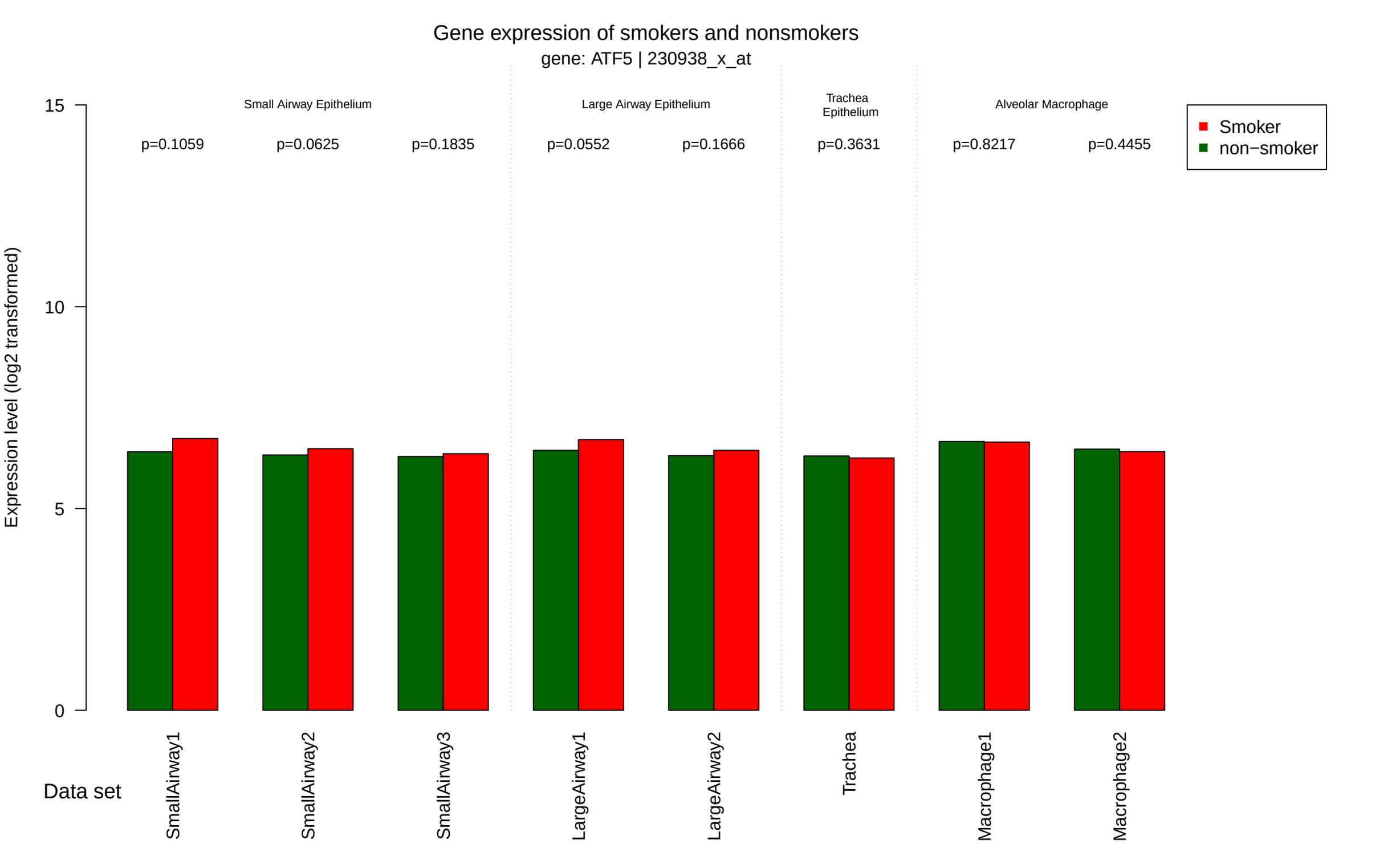
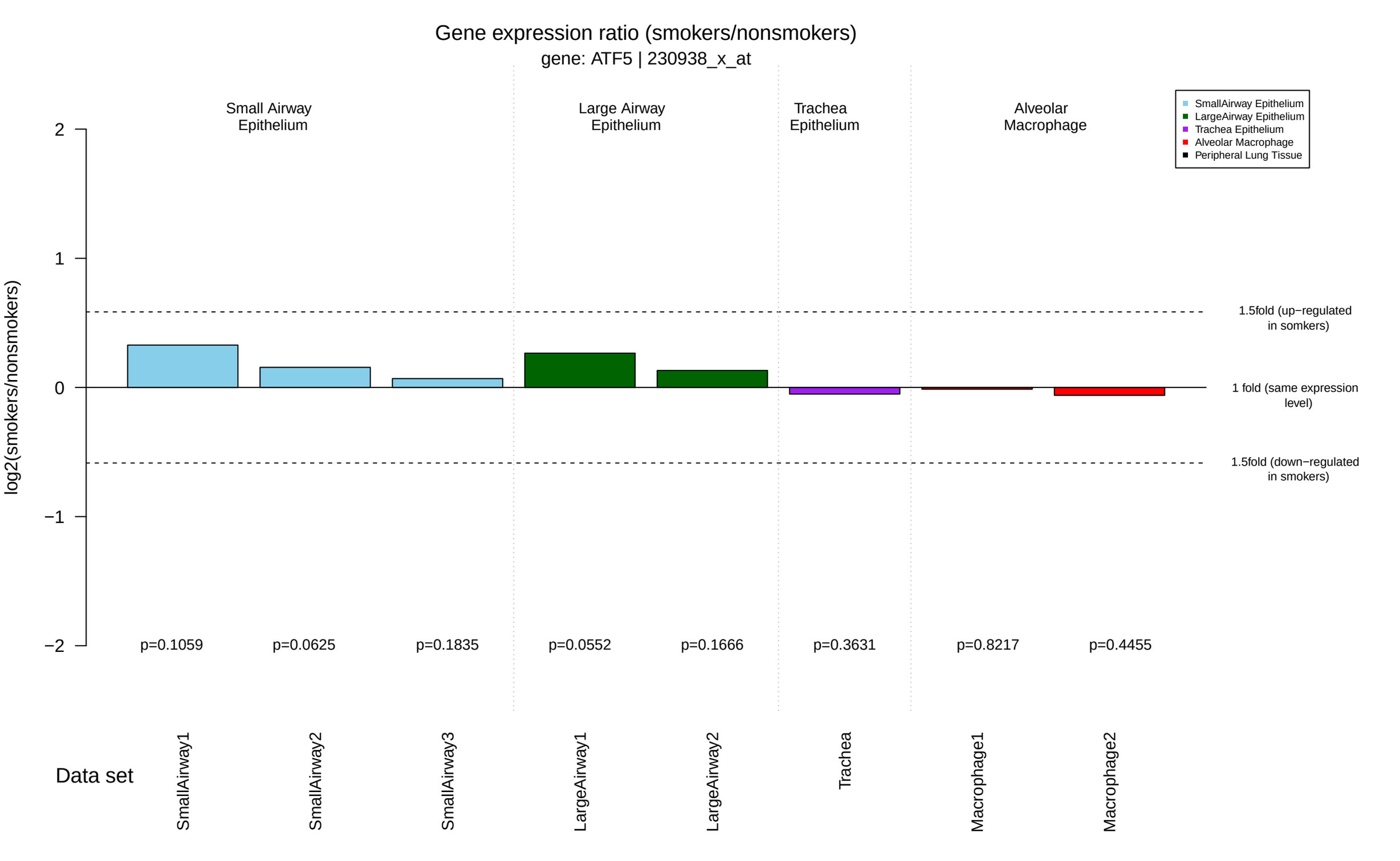
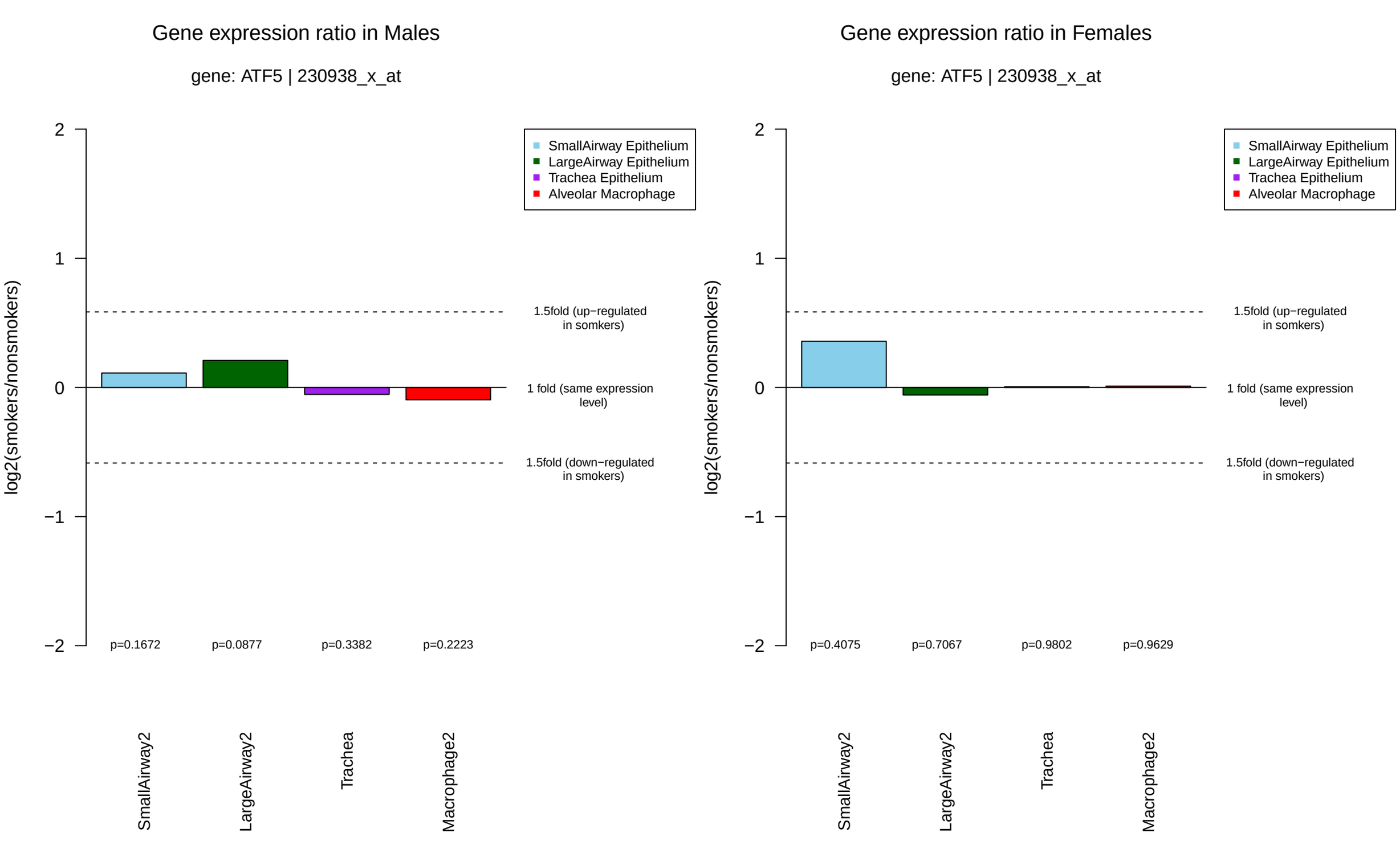
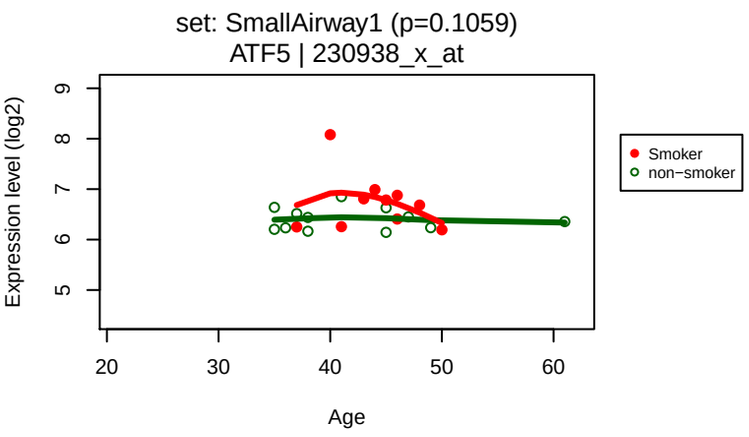
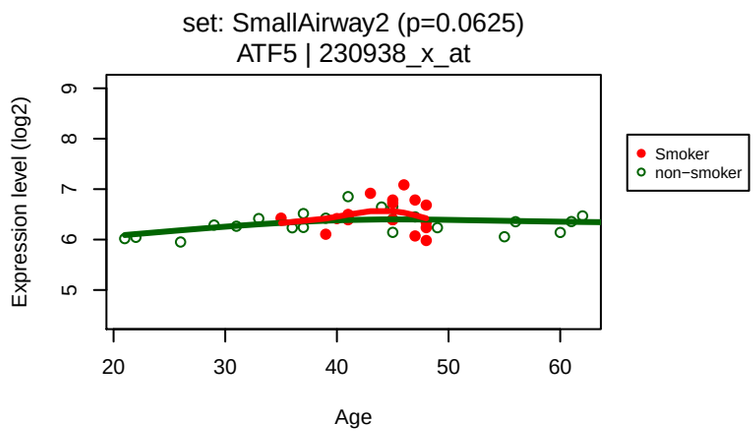
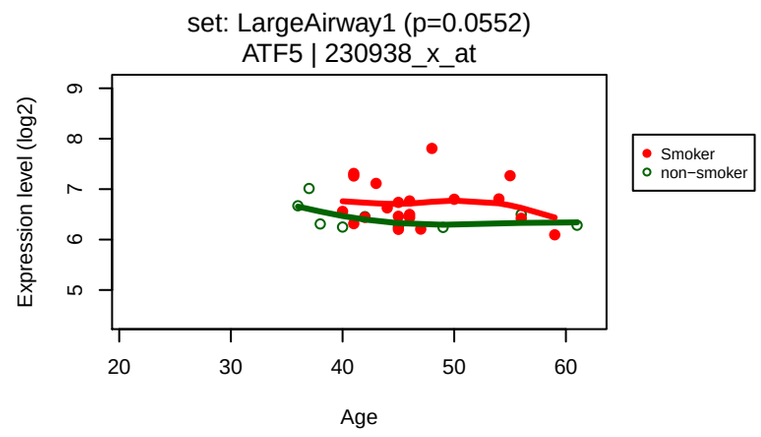

Gene expression of smokers and nonsmokers
gene: ATF5 | 230938_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1059
p=0.0625
p=0.1835
p=0.0552
p=0.1666
p=0.3631
p=0.8217
p=0.4455
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

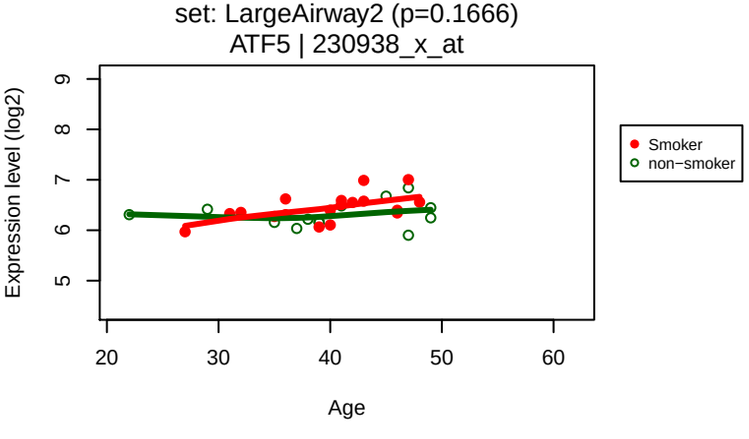

Gene expression ratio (smokers/nonsmokers)
gene: ATF5 | 230938_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1059
p=0.0625
p=0.1835
p=0.0552
p=0.1666
p=0.3631
p=0.8217
p=0.4455
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

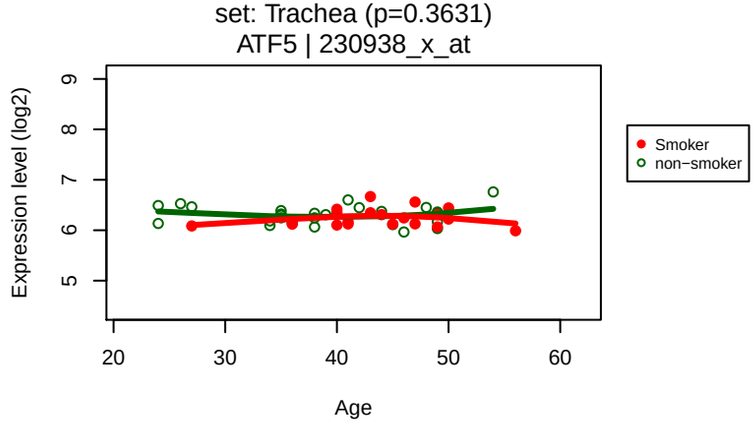

Gene expression ratio in Males
gene: ATF5 | 230938_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1672
p=0.0877
p=0.3382
p=0.2223
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATF5 | 230938_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4075
p=0.7067
p=0.9802
p=0.9629
SmallAirway2
LargeAirway2
Trachea
Macrophage2

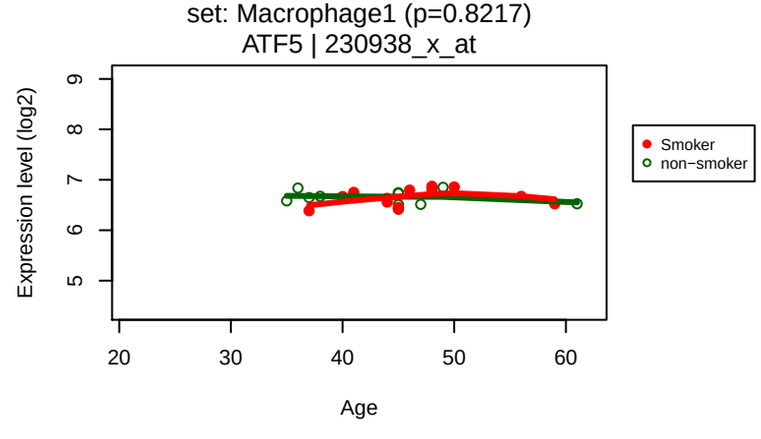

set: SmallAirway1 (p=0.1059)
ATF5 | 230938_x_at
Expression level (log2)
Age
Smoker
non−smoker

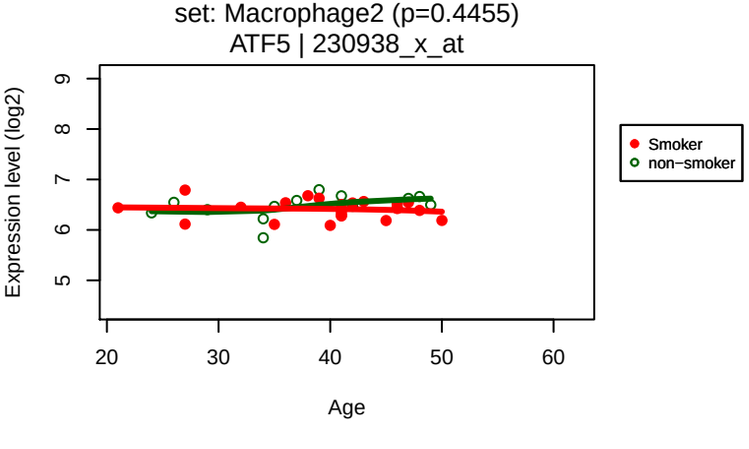

set: SmallAirway2 (p=0.0625)
ATF5 | 230938_x_at
Expression level (log2)
Age
Smoker
non−smoker

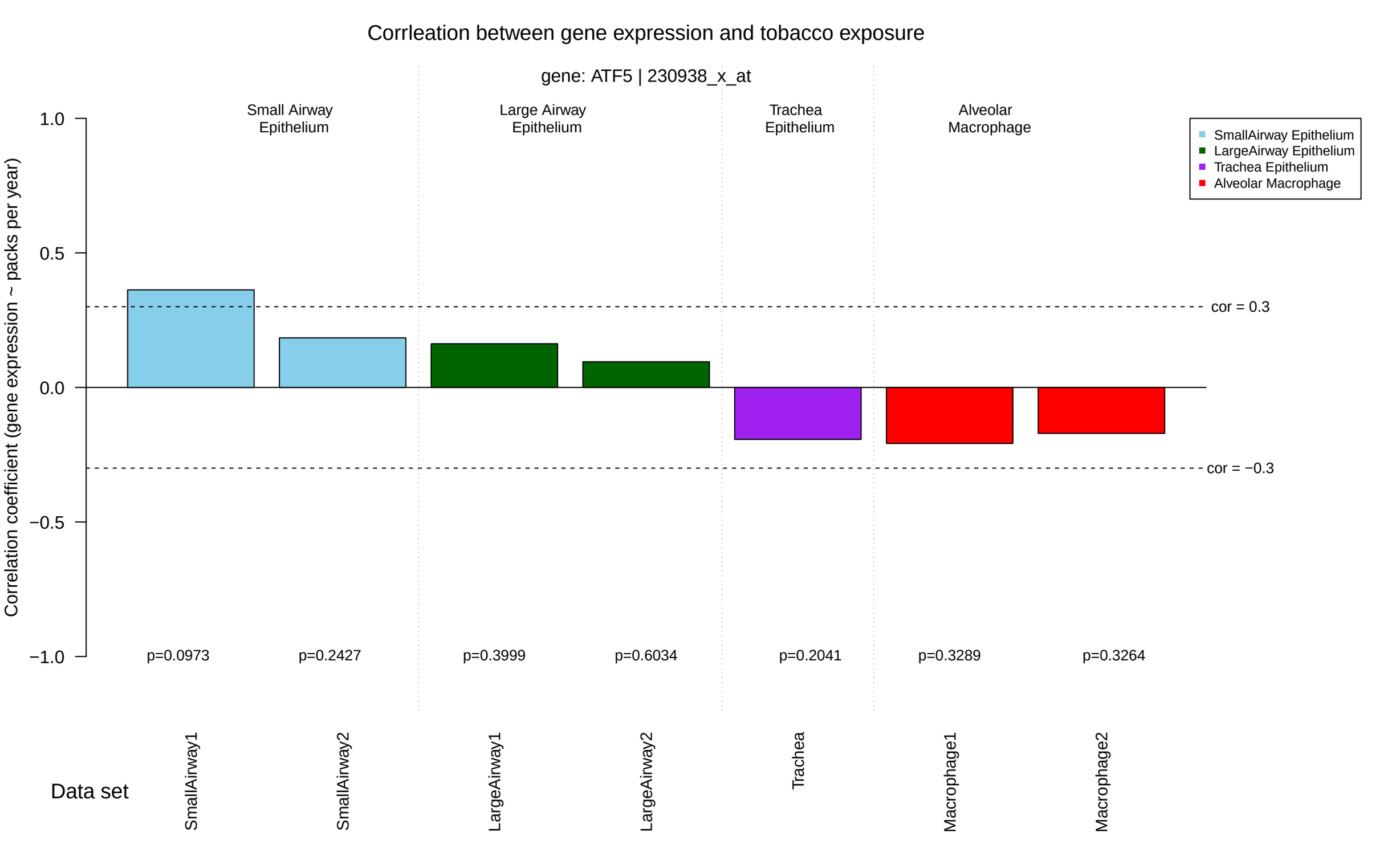

set: LargeAirway1 (p=0.0552)
ATF5 | 230938_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1666)
ATF5 | 230938_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3631)
ATF5 | 230938_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8217)
ATF5 | 230938_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4455)
ATF5 | 230938_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATF5 | 230938_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0973
p=0.2427
p=0.3999
p=0.6034
p=0.2041
p=0.3289
p=0.3264
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2